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Nano thermal diode based on polarization control of nanoparticles on graphene nanograting

Abstract

A thermal diode includes: a dielectric enclosure having a top surface, a bottom surface, a first side, and a second side; a first Weyl semimetal nanoparticle disposed on the first side; a second Weyl semimetal nanoparticle disposed on the second side; a nanograting disposed on the bottom surface; and a voltage source configured to provide a voltage bias to the nanograting to suppress surface waves from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and to modify surface waves from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle.

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Background/Summary

TECHNICAL FIELD

(1) One or more embodiments relate generally to thermal diodes.

Description

BACKGROUND

(1) A Weyl semimetal (WS) is a solid state crystal, whose low energy excitations are Weyl fermions that carry electrical charge even at room temperatures.

(2) FIG. 1A illustrates a related art thermal diode using InSb in one mode of operation.

(3) As shown in the figure, a InSb particle **104** and an InSb particle **106** are suspended above a bulk InSb substrate **102**. In this example, InSb particle **104** is heated to a much higher temperature than that of InSb particle **106**. As a result, InSb particle **104** will transfer heat to InSb particle **106** as shown by arrow **110**. This net heat transfer is termed heat flux, $Q_{sub.12}$, which is the difference between the heat emitted by particle **104** and the heat received by particle **104** due to the thermal emission by particle **106** and wherein the “heat” is in the form of near field thermal radiation and the flux is in a direction as indicated by arrow **110**.

- (4) FIG. 1B illustrates the related art thermal diode of FIG. 1A in a second mode of operation, wherein InSb particle **106** is heated to a much higher temperature than that of InSb particle **104**. As a result, InSb particle **106** will transfer heat to InSb particle **104** as shown by arrow **112**. This net heat transfer is termed heat flux, $Q_{sub.21}$, which is the difference between the heat emitted by particle **106** and the heat received by particle **106** due to the thermal emission by particle **104** and wherein the “heat” is in the form of near field thermal radiation and the flux is in a direction as indicated by arrow **112**.
- (5) In the systems of FIGS. 1A-B, a strong magnetic field **108** is provided, wherein magnetic field **108** is a direction into the figure, and is on the order of 2 T.
- (6) In operation, by applying magnetic field **108**, the electromagnetic waves propagating at the surface of substrate **102** becomes nonreciprocal, where at certain frequencies the waves traveling in one direction at the surface is supported whereas that in the opposite direction is suppressed.
- (7) It should be noted that atoms at the surface of substrate **102** may still act similarly with respect to electromagnetic waves propagating in one direction verses another direction. However, the microscopic structure inside substrate **102** may be different when a magnetic field is applied. In particular, in the presence of a magnetic field, off-diagonal components appear in the permittivity tensor and the electromagnetic wave transport becomes asymmetric at the surface of substrate **102**. Mathematically, it may be described as the solution to the wave function at the surface of substrate **102** has solutions in one direction, while in the other direction there is no solution.
- (8) Thus, the net heat fluxes in two operation modes, i.e., $Q_{sub.12}$ as shown in FIG. 1A and $Q_{sub.21}$ as shown in FIG. 1B, will be different. In other words, heat flows more in one direction as compared to the other direction, which enables the thermal rectification.
- (9) However, the system discussed in FIGS. 1A-B requires strong magnetic fields to realize a large rectification ratio. When the magnetic field is 2 T, the maximum rectification ratio reaches approximately 6. This will be described in greater detail with reference to FIG. 2.
- (10) FIG. 2 illustrates a rectification graph **200** of the related art thermal diode of FIGS. 1A-B. As shown in FIG. 2, graph **200** includes a y-axis **202**, an x-axis **204**, and a function **206**.
- (11) Y-axis **202** corresponds to a rectification ratio, η , defined as follows:
- (12)
$$\eta = (Q_{12} - Q_{21}) / Q_{12} \quad (1)$$
- (13) X-axis **204** corresponds to the distance between InSb particle **104** and InSb particle **106**, measured in μm .
- (14) The maximum rectification is noted at point **208** of function **206**, whereas dotted line **210** represents an asymptotic limit of the rectification, which is equal to 1 and indicates that no rectification occurs.
- (15) In addition to the large magnetic field needed, another problem with the system discussed above with reference to FIGS. 1A-B is that suspending InSb particle **104** and InSb particle **106** above substrate **102** is practically challenging. Moreover, to tune the rectification ratio, the distance between InSb particle **104** and InSb particle **106** needs to be changed, which again would be difficult to achieve in reality.
- (16) What is needed is a thermal diode that may achieve improved rectification over that of the system discussed above with reference to FIGS. 1A-2, without the need for suspending particles over a substrate and without the need of such a large magnetic field.

SUMMARY

- (17) An aspect of the present disclosure is drawn to a thermal diode including: a dielectric enclosure having a top surface, a bottom surface, a first side, and a second side; a first Weyl semimetal nanoparticle disposed on the first side; a second Weyl semimetal nanoparticle disposed on the second side; a nanograting disposed on the bottom surface; and a voltage source configured to provide a voltage bias to the nanograting to suppress surface waves from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and to modify surface waves

from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle.

(18) In some embodiments of this aspect, the first Weyl semimetal nanoparticle and the second Weyl semimetal nanoparticle comprise a material selected from the group of materials consisting of Co.sub.3Sn.sub.2S.sub.2, Mn.sub.3Sn, EuCd.sub.2As.sub.2, TaAs, MoTe.sub.2, NbAs, YbMnBi.sub.2, and WTe.sub.2.

(19) In some embodiments of this aspect, the dielectric enclosure includes SiO.sub.2.

(20) In some embodiments of this aspect, the distance between the nanograting and the first Weyl semimetal nanoparticle is 80 nm.

(21) In some embodiments of this aspect, the nanograting includes graphene.

(22) Another aspect of the present disclosure is drawn to a method of making a thermal diode, the method including: providing a nanograting; forming a first portion of a dielectric on top of the provided nanograting; depositing a first Weyl semimetal nanoparticle at a first location on top of the first portion of the dielectric; depositing a second Weyl semimetal nanoparticle at a second location on top of the first portion of the dielectric; forming a second portion of the dielectric on top of the first portion of the dielectric, the first Weyl semimetal nanoparticle, and the second Weyl semimetal nanoparticle; and connecting a voltage source to the nanograting, wherein the voltage source is configured to provide a voltage bias to the nanograting to suppress surface waves from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and to modify surface waves from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle

(23) In some embodiments of this aspect, the depositing of the first Weyl semimetal nanoparticle at the first location on top of the first portion of the dielectric includes depositing the first Weyl semimetal nanoparticle including a material selected from the group of materials consisting of Co.sub.3Sn.sub.2S.sub.2, Mn.sub.3Sn, EuCd.sub.2As.sub.2, TaAs, MoTe.sub.2, NbAs, YbMnBi.sub.2, and WTe.sub.2.

(24) In some embodiments of this aspect, the forming of the first portion of the dielectric on top of the provided nanograting includes forming the first portion of the dielectric including SiO.sub.2.

(25) In some embodiments of this aspect, the depositing of the second Weyl semimetal nanoparticle at the second location on top of the first portion of the dielectric includes depositing the second Weyl semimetal nanoparticle from the first Weyl semimetal nanoparticle of 1 μm .

(26) In some embodiments of this aspect, the providing of the nanograting includes providing a graphene nanograting.

(27) Another aspect of the present disclosure is drawn to a method including: providing a thermal diode including: a dielectric enclosure having a top surface, a bottom surface, a first side, and a second side; a first Weyl semimetal nanoparticle disposed on the first side; a second Weyl semimetal nanoparticle disposed on the second side; a nanograting disposed on the bottom surface; and a voltage source configured to provide a voltage bias to the nanograting to suppress surface waves from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and to modify surface waves from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle; applying the voltage bias to the nanograting; and rotating the first Weyl semimetal nanoparticle and the second Weyl semimetal nanoparticle about an axis by an angle Θ , wherein $0^\circ \leq \Theta \leq 360^\circ$.

(28) In some embodiments of this aspect, the providing of the thermal diode includes providing the thermal diode such that the first Weyl semimetal nanoparticle and the second Weyl semimetal nanoparticle comprise a material selected from the group of materials consisting of Co.sub.3Sn.sub.2S.sub.2, Mn.sub.3Sn, EuCd.sub.2As.sub.2, TaAs, MoTe.sub.2, NbAs, YbMnBi.sub.2, and WTe.sub.2.

(29) In some embodiments of this aspect, the providing of the thermal diode includes providing the thermal diode such that the dielectric enclosure includes SiO.sub.2.

(30) In some embodiments of this aspect, the providing of the thermal diode includes providing the

thermal diode such that the distance between the nanograting and the first Weyl semimetal nanoparticle is 80 nm.

(31) In some embodiments of this aspect, the providing of the thermal diode includes providing the thermal diode such that the nanograting includes graphene.

(32) In some embodiments of this aspect, the method further includes applying, via the voltage source, a second voltage bias to the nanograting; and rotating the second Weyl semimetal nanoparticle about the axis by a second angle $\Theta_{sub.2}$, wherein the voltage source is additionally configured to provide the second voltage bias to the nanograting to suppress surface waves from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle and to modify surface waves from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle, wherein a first net heat flux is different from a second net heat flux, wherein the first net heat flux corresponds to a difference between a first heat flux from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and a second heat flux from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle when the second Weyl semimetal nanoparticle is rotated about the axis by the angle Θ , wherein the second net heat flux corresponds to a difference between a third heat flux from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and a fourth heat flux from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle when the second Weyl semimetal nanoparticle is rotated about the axis by the angle $\Theta_{sub.2}$, wherein $0^\circ \leq \Theta_{sub.2} \leq 360^\circ$, and wherein $\Theta \neq \Theta_{sub.2}$.

(33) In some embodiments of this aspect, the rotating of the second Weyl semimetal nanoparticle about the axis by the angle Θ , includes applying a magnetic field to the second Weyl semimetal nanoparticle. In some of these embodiments, the applying of the magnetic field to the second Weyl semimetal nanoparticle includes applying the magnetic field to the second Weyl semimetal nanoparticle for a predetermined time period. In some of these embodiments, the applying of the magnetic field to the second Weyl semimetal nanoparticle includes applying the magnetic field at a predetermined field strength to the second Weyl semimetal nanoparticle.

BRIEF SUMMARY OF THE DRAWINGS

(34) The accompanying drawings, which are incorporated in and form a part of the specification, illustrate and explain example embodiments. In the drawings:

(35) FIG. 1A illustrates a related art thermal diode in one mode of operation;

(36) FIG. 1B illustrates the related art thermal diode of FIG. 1A in a second mode of operation;

(37) FIG. 2 illustrates a rectification graph of the related art thermal diode of FIGS. 1A-B;

(38) FIG. 3 illustrates a thermal diode system in accordance with aspects of the present disclosure;

(39) FIG. 4 illustrates an example method of making a thermal diode in accordance with aspects of the present disclosure;

(40) FIG. 5A-H illustrate eight processes, respectively, for making a thermal diode in accordance with aspects of the present disclosure;

(41) FIG. 6 illustrates a method of operating a thermal diode in accordance with aspects of the present disclosure;

(42) FIG. 7 illustrates a graph of heat flux as a function of rotation angle of a Weyl semimetal nanoparticle having one loss factor in two modes of operation in accordance with aspects of the present disclosure;

(43) FIG. 8 illustrates a graph of a ratio of heat flux as a function of rotation angle of two Weyl semimetal nanoparticles having one loss factor;

(44) FIG. 9 illustrates another graph of heat flux as a function of rotation angle of two Weyl semimetal nanoparticles having another loss factor in two modes of operation in accordance with aspects of the present disclosure;

(45) FIG. 10 illustrates a graph of a ratio of heat flux as a function of rotation angle of two Weyl semimetal nanoparticles having another loss factor in two modes of operation of FIG. 9; and

(46) FIGS. 11A-D illustrate another example thermal diode in accordance with aspects of the present disclosure, being operated in four different modes, respectively.

DETAILED DESCRIPTION

(47) A thermal diode in accordance with aspects of the present disclosure achieves improved rectification over that of the system discussed above with reference to FIGS. 1A-2, without the need for suspending nanoparticles over a substrate and without the need of applying a large external magnetic field.

(48) A nanoscale thermal diode that includes Weyl semimetal nanoparticles and a graphene nanograting in accordance with aspects of the present disclosure achieves a thermal rectification ratio ranging from $10^{\sup.-2}$ to $2 \times 10^{\sup.3}$ by actively controlling the nonreciprocal surface waves in graphene nanograting, the polarization and the material property of the Weyl semimetal nanoparticles.

(49) In accordance with aspects of the present disclosure, WS particles are embedded in a dielectric enclosure to position them at desired distance above a graphene nanograting. The graphene nanograting provides much stronger nonreciprocal response when a bias voltage is applied. A tunable thermal rectification can be realized by changing the polarization of the WS particles, such as exciting the dipole response of the particles with differently polarized electromagnetic waves, which does not require changing the geometry of either particles or graphene nanograting. The thermal rectification characteristic can be additionally tuned by changing the loss factor of the WS, which can be practically implemented by introducing appropriate dopants or properly choosing the WS materials.

(50) An example thermal diode, method of making and method of operating the same in accordance with aspects of the present disclosure will now be described in greater detail with reference to FIGS. 3-11D.

(51) FIG. 3 illustrates a thermal diode system **300** in accordance with aspects of the present disclosure.

(52) As shown in the figure, thermal diode system **300** includes a dielectric enclosure **302**, a nanograting **304**, a voltage source **306**, a WS nanoparticle (WSN) **308**, a WSN **310**, a magnetic field source **312**, a conducting line **314**, and a conducting line **316**.

(53) Dielectric enclosure **302** may include any commonly used dielectric materials, a non-limiting example of which includes SiO_2 . Dopant may include Boron and Phosphorous, or any known p or n type dopants. The amount of doping, for example on the order of $10^{\sup.15} \sim 10^{\sup.19} \text{ cm}^{\sup.-3}$, as will be described in greater detail below, affects the amount of thermal rectification.

(54) Magnetic field source **312** is configured to provide a magnetic field in a direction indicated by line **318** to affect WS particle **310**, as will be described in more detail below. Magnetic field source **312** may be any known magnetic field source, non-limiting examples of which include permanent magnets and electromagnets.

(55) An x-axis **320**, a y-axis **322**, and a z-axis **324** are shown for reference.

(56) WSN **308** and WSN **310** are embedded within dielectric enclosure **302**. WSN **308** is separated from WSN **310** by a predetermined distance along y-axis **322**. In a non-limiting example embodiment, WSN **308** is separated from WSN **310** by 1 μm .

(57) WSN **308** and WSN **310** may be any known WS, non-limiting examples of which include $\text{Co}_{\text{sub}.3}\text{Sn}_{\text{sub}.2}\text{S}_{\text{sub}.2}$, $\text{Mn}_{\text{sub}.3}\text{Sn}$, $\text{EuCd}_{\text{sub}.2}\text{As}_{\text{sub}.2}$, TaAs , $\text{MoTe}_{\text{sub}.2}$, NbAs , $\text{YbMnBi}_{\text{sub}.2}$, and $\text{WTe}_{\text{sub}.2}$. In an example embodiment, each of WSN **308** and WSN **310** may have a radius in the range of 5-30 nm.

(58) In an example embodiment, nanograting **304** is selected from the group of graphene and borophene. In an example embodiment, the distance from nanograting **304** to each of WSN **308** and WSN **310** in a direction of z-axis **324** is within a range of 80-250 nm. Nanograting **304** has a periodicity of $p+w$, wherein in a non-limiting example embodiment $p+w=10 \text{ nm}$ and a width of $p=5 \text{ nm}$.

(59) A dipolar response of a WSN can be externally excited by an electromagnetic wave with a preferred polarization. Such polarization may be defined by an angle in the xz-plane about the y-axis. The polarization-dependent excitation is equivalent to implementing a rotation of the nanoparticles around the y-axis by an angle Θ . When voltage source **306** provides a bias voltage $V_{sub,d}$ by way of conducting line **314** to nanograting **304**, the electromagnetic waves propagating along the y-axis will exhibit strong direction preference.

(60) For example, the surface wave from WSN **308** to WSN **310** may be suppressed, while the opposite wave may be modified to be enhanced. Such directional preference will enable selectively propagating heat energy between WSN **308** and WSN **310**, which realizes the thermal diode.

(61) In other words, a voltage bias $V_{sub,d}$ applied to nanograting **304** creates an electric field within dielectric enclosure **302**, which affects surface waves within dielectric enclosure **302**. The magnetic field from magnetic field source **312** rotates WSN **310**. In some embodiments, the rotation of WSN **310** may be accomplished when turning on the magnetic field to apply a magnetic force. Once the rotation is completed, the magnetic field may be turned off. The effect of the rotating of WSN **310** and the surface waves is such that, near field thermal radiation as transmitted from WSN **308** to WSN **310**, when WSN **308** is of a higher temperature as that of WSN **310**, is different from the near field thermal radiation as transmitted from WSN **310** to WSN **308**, when WSN **310** is of a higher temperature as that of WSN **308**. This difference in the directional flow of the near field thermal radiation defines the thermal diode.

(62) FIG. **4** illustrates an example method **400** of making a thermal diode in accordance with aspects of the present disclosure. As shown in the figure, method **400** starts (S**402**) and a nanograting is provided (S**404**). This will be described in greater detail with reference to FIGS. 5A-C.

(63) FIG. 5A-H illustrate eight processes, respectively, for making a thermal diode in accordance with aspects of the present disclosure.

(64) As shown in FIG. 5A, a layer **502** of graphene is formed. This may be performed by any known method, a non-limiting example of which includes chemical vapor deposition.

(65) As shown in FIG. 5B, layer **502** of graphene is etched to create a grating. This may be performed by any known method, a non-limiting example of which includes etching via mask. In this example, layer **502** of graphene is etched to provide graphene portions **504**, **506**, **508** and **510**, separated by spacings **512**, **514**, and **516**.

(66) As shown in FIG. 5C, a filler material, such as polymethyl methacrylate (PMMA), is spin coated into spacings **512**, **514** and **516** to as filler area **518**, **520**, and **522** to provide a flat surface to grow a dielectric layer and to form a mechanical support, as will be described below.

(67) Returning to FIG. **4**, after the nanograting is provided (S**404**), a dielectric layer is formed (S**406**). This will be described in greater detail with reference to FIG. 5D.

(68) As shown in FIG. 5D, a dielectric layer **524** is deposited or thermally grown onto the graphene portions and filler portions. Dielectric layer **524** may deposited by known depositions methods, such as for example electron beam deposition or sputtering, and may be doped during this process by known doping methods.

(69) Returning to FIG. **4**, after a dielectric layer is formed (S**406**), the WS nanoparticles are deposited (S**408**). This will be described in greater detail with reference to FIG. 5E.

(70) As shown in FIG. 5E, WSP **308** and WSP **310** (not shown) may be deposited on dielectric layer **524**, by a micro-manipulator equipped with scanning electron microscopy.

(71) The precise control of the position where WSP **308** and WSP **310** are deposited may be achieved by using a micro-manipulator equipped with scanning electron microscopy (SEM), wherein the micro-manipulator has a nanoscale tip that can move the nanoparticles.

(72) Returning to FIG. **4**, after the Weyl semimetal nanoparticles are deposited (S**408**), the remainder of the dielectric is formed (S**410**). This will be described in greater detail with reference to FIG. 5F.

(73) As shown in FIG. 5F, the same dielectric that was deposited prior, as discussed above with reference to FIG. 5D, is again deposited to embed WSP **308** and WSP **310** (not shown). By using the same dielectric, during this process, lattice mismatch is minimized to improve the structure of resulting dielectric enclosure **302**.

(74) Returning to FIG. 4, after the remainder of the dielectric is formed (**S410**), the voltage source is connected (**S412**). This will be described in greater detail with reference to FIGS. 5G-H.

(75) As shown in FIG. 5G, filler areas **518**, **520**, and **522** are removed via etching with acetone.

(76) As shown in FIG. 5H, voltage source **306** is connected to nanograting **304** via conducting line **314** and conducting line **316**.

(77) Returning to FIG. 4, after the voltage source is connected (**S412**), method **400** stops (**S414**).

(78) Now that a thermal diode in accordance with aspects of the present disclosure has been created, it may be operated.

(79) FIG. 6 illustrates a method **600** of operating a thermal diode in accordance with aspects of the present disclosure.

(80) As shown in the figure, method **600** starts (**S602**) and a thermal diode is provided (**S604**). For example, thermal diode system **300** may be provided as described above with reference to FIGS. 3-5H.

(81) Returning to FIG. 6, after the thermal diode is provided (**S604**), a bias voltage is applied (**S606**). For example, as shown in FIG. 3, voltage source **306** may provide a voltage bias to nanograting **304** via conducting lines **314** and **316**. In non-limiting example embodiments, the voltage bias is in a range from -20 V to 20 V .

(82) Returning to FIG. 6, after a bias voltage is applied (**S606**), both WSNs are rotated (**S608**). For example, as shown in FIG. 3, magnetic field source **312** generates a magnetic field that rotates WSN **308** and WSN **310** by the same predetermined angle Θ . The angle Θ may be determined from a priori data developed from previous testing of thermal diode system **300**. This will be described in greater detail with reference to FIGS. 7-10.

(83) FIG. 7 illustrates a graph **700** of heat flux as a function of rotation angle of a Weyl semimetal nanoparticle in two modes of operation with a loss factor τ of 0.01 femto-seconds (fs) in accordance with aspects of the present disclosure. The loss factor τ of 0.01 fs has been experimentally verified in the literature for the WS materials, though practically other loss factors may be chosen by appropriately doping the WS nanoparticles.

(84) As shown in the figure, graph **700** has a y-axis **702** of heat flux measured in units of W/m^2 , an x-axis **704** of rotation angle Θ measured in radians, a plurality of data points **706**, and a plurality of data points **708**.

(85) X-axis **704** corresponds to the angle Θ for which WSN **308** and WSN **310** are rotated. The plurality of data points **706** correspond to the heat flux in a direction from WSN **308** to WSN **310**. The plurality of data points **708** correspond to the heat flux in a direction from WSN **310** to WSN **308**.

(86) It is clear from graph **700** that the heat flux in a direction from WSN **308** to WSN **310** associated with the plurality of data points **706** is on the order of 10-100 times larger than the heat flux in a direction from WSN **310** to WSN **308** associated with the plurality of data points **708**. This holds true for $0 \leq \Theta \leq \pi$.

(87) FIG. 8 illustrates a graph **800** of a ratio of heat flux as a function of rotation angle of two WSNs in two modes of operation of FIG. 7.

(88) As shown in the figure, graph **800** has a y-axis **802** of a ratio of heat flux, an x-axis **804** of rotation angle Θ measured in radians, and a plurality of data points **806**.

(89) The ratio of heat flux of y-axis **802** is a ratio of the heat flux in a direction from WSN **310** to WSN **308** over the heat flux in a direction from WSN **308** to WSN **310**.

(90) It should be noted that at $\Theta=0$ radians, the heat flux in the direction from WSN **308** to WSN **310**, $Q_{\text{sub.21}}$, is on the order of about 34 times larger than the heat flux in the direction from WSN

310 to WSN **308**, Q.sub.12. It should be noted that 34 times larger is a much more pronounced thermal diode as compared with the 6-fold increase as discussed with the related art above with reference to FIG. 2.

(91) FIG. 9 illustrates another graph **900** of heat flux as a function of rotation angle of two WSNs in two modes of operation with a loss factor τ of 1.0 femto-seconds (fs) in accordance with aspects of the present disclosure.

(92) As shown in the figure, graph **900** has a y-axis **902** of heat flux measured in units of W/m.sup.2, an x-axis **904** of rotation angle Θ measured in radians, a plurality of data points **906**, and a plurality of data points **908**.

(93) X-axis **905** corresponds to the angle Θ for which WSN **308** and WSN **310** are rotated. The plurality of data points **906** correspond to the heat flux in a direction from WSN **308** to WSN **310**. The plurality of data points **908** correspond to the heat flux in a direction from WSN **310** to WSN **308**.

(94) It should be noted that as compared with graph **700** of FIG. 7, in this case, the total heat flux changes drastically over range of rotation angle Θ . In particular, at $\Theta=0$ radians, the heat flux from WSN **308** to WSN **310**, i.e., Q.sub.12, as shown by dot **910** is on the order of 10.sup.2 larger than the heat flux from WSN **310** to WSN **308**, i.e., Q.sub.21, as shown by triangle **912**.

(95) Further, at $\Theta \neq \pi/3$ radians, the heat flux from WSN **308** to WSN **310**, i.e., Q.sub.12, as shown by area **914** is roughly equal to the heat flux from WSN **310** to WSN **308**, i.e., Q.sub.21.

(96) Finally, at $\Theta=\pi$ radians, the heat flux from WSN **308** to WSN **310**, i.e., Q.sub.12, as shown by dot **916** is on the order of 10.sup.3 smaller than the heat flux from WSN **310** to WSN **308**, i.e., Q.sub.21, as shown by triangle **918**. Therefore, in this embodiment, wherein the doping in WSNs is so much higher as to increase the loss factor from 0.1 fs, as discussed above with reference to FIGS. 7-8, to a loss factor of 1.0 fs, the heat flux response is able to transition from one direction to a second direction based on the rotation angle of the WSN **310**.

(97) FIG. 10 illustrates a graph **1000** of a ratio of heat flux as a function of rotation angle of two WSNs in two modes of operation of FIG. 9.

(98) As shown in the figure, graph **1000** has a y-axis **1002** of a ratio of heat flux, an x-axis **1004** of rotation angle Θ measured in radians, and a plurality of data points **1006**.

(99) The ratio of heat flux of y-axis **1002** is a ratio of the heat flux in a direction from WSN **310** to WSN **308** over the heat flux in a direction from WSN **308** to WSN **310**. As shown in the figure, the change in heat flux between $0 \leq \Theta \leq \pi$ is on the order of 10.sup.5 (10.sup.3-10.sup.-2).

(100) Returning to FIG. 6, after the WSNs are rotated (S608), method **600** stops (S610).

(101) In the above-discussed non-limiting embodiments, **306** provided a voltage bias to nanograting **304** to effect of the surface waves the of dielectric enclosure **302**. With this in mind, in accordance with aspects of the present disclosure, the surface waves of the dielectric enclosure **302** may be further modified to tune the thermal diode system by modifying the voltage bias applied to nanograting **304**. This will be described in greater detail with reference to FIGS. 11A-D.

(102) FIGS. 11A-D illustrate another example thermal diode system **1100** in accordance with aspects of the present disclosure, being operated in four different modes, respectively.

(103) As shown in FIG. 11A, thermal diode system **1100** includes dielectric enclosure **302**, nanograting **304**, voltage source **306**, WSN **308** and a voltage controller **1102**. Nanograting **304** includes graphene portions **1104**, **1106**, **1108**, and **1110**. Voltage controller **1102** is configured to provide a positive voltage from voltage source **306** to one of: graphene portion **1104** via a conducting line **1112**; to graphene portion **1106** via a conducting line **1114**; to graphene portion **1108** via a conducting line **1116**, and to graphene portion **1110** via a conducting line **1118**. Voltage controller **1102** is configured to additionally to return a negative voltage to voltage source **306** from one of: graphene portion **1104** via a conducting line **1120**; graphene portion **1106** via a conducting line **1122**; to graphene portion **1108** via a conducting line **1124**, and to graphene portion **1110** via a conducting line **1126**.

(104) Voltage controller **1102** is configured to communicate with graphene portions **1104**, **1106**, **1108**, and **1110** via conducting lines **1112**, **1114**, **1116**, and **1118** respectively. Further, voltage controller **1102** is configured to communicate with **1104**, **1106**, **1108**, and **1110** via conducting lines **1112**, **1114**, **1116**, and **1118** respectively.

(105) Voltage source **306** is configured to provide a positive voltage to graphene portion **1104** via conducting line **1112** and to receive a negative voltage from graphene portion **1110** via the conducting line **1126**. By providing a voltage to each of graphene portions **1104** and **1110**, an electric field is generated between graphene portion **1104** and graphene portion **1110** as indicted by dotted arrows **1128**.

(106) The electric field generated between graphene portion **1104** and graphene portion **1110** as indicted by dotted arrows **1128** will have a predetermined influence on both the heat flux from WSN **308** to WSN **310**, i.e., Q.sub.12, and the heat flux from WSN **310** to WSN **308**, i.e., Q.sub.21.

(107) As shown in FIG. **11B**, the thermal diode system **1100** is operated in a manner opposite to that as discussed above with reference to FIG. **11A**.

(108) In particular, in FIG. **11B**, voltage source **306** is configured to provide a negative voltage to graphene portion **1104** via conducting line **1112** and to receive a positive voltage from graphene portion **1110** via the conducting line **1126**. By providing a voltage to each of graphene portions **1104** and **1110**, an electric field is generated between graphene portion **1104** and graphene portion **1110** as indicted by dotted arrows **1130**.

(109) The electric field generated between graphene portion **1104** and graphene portion **1110** as indicted by dotted arrows **1130** will have a predetermined influence on both the heat flux from WSN **308** to WSN **310**, i.e., Q.sub.12, and the heat flux from WSN **310** to WSN **308**, i.e., Q.sub.21, wherein the influence is different from that as discussed above with reference to FIG. **11A**.

(110) It should be noted that an electric field need not be generated from the outer bounds of nanograting **304**. On the contrary, in accordance with aspects of the present disclosure, any combination of portions of nanograting **304** may be activated to affect the surface waves within dielectric enclosure **302**. Some example embodiments of other portions of nanograting **304** being activated to affect the surface waves within dielectric enclosure **302** will now be described with reference to FIGS. **11C-D**.

(111) As shown in FIG. **11C**, voltage controller **1102** is configured to provide a positive voltage from voltage source **306** to graphene portion **1106** via a conducting line **1114**. Voltage controller **1102** is configured to additionally to return a negative voltage to voltage source **306** from graphene portion **1108** via a conducting line **1124**. By providing a voltage to each of graphene portions **1106** and **1108**, an electric field is generated between graphene portion **1106** and graphene portion **1108** as indicted by dotted arrows **1132**.

(112) The electric field generated between graphene portion **1106** and graphene portion **1108** as indicted by dotted arrows **1132** will have a predetermined influence on both the heat flux from WSN **308** to WSN **310**, i.e., Q.sub.12, and the heat flux from WSN **310** to WSN **308**, i.e., Q.sub.21. This influence may be different that that as discussed above with reference to FIG. **11A**.

(113) As shown in FIG. **11D**, the thermal diode system **1100** is operated in a manner opposite to that as discussed above with reference to FIG. **11C**.

(114) In particular, in FIG. **11D**, voltage source **306** is configured to provide a negative voltage to graphene portion **1106** via conducting line **1114** and to receive a positive voltage from graphene portion **1108** via the conducting line **1122**. By providing a voltage to each of graphene portions **1106** and **1108**, an electric field is generated between graphene portion **1106** and graphene portion **1108** as indicted by dotted arrows **1134**.

(115) The electric field generated between graphene portion **1106** and graphene portion **1108** as indicted by dotted arrows **1134** will have a predetermined influence on both the heat flux from

WSN 308 to WSN 310, i.e., Q.sub.12, and the heat flux from WSN 310 to WSN 308, i.e., Q.sub.21, wherein the influence is different from that as discussed above with reference to FIG. 11C.

(116) In light of the above discussion with respect to FIGS. 11A-D, it should be noted that any combination of any number of graphene portions may be excited to produce a thermal diode in accordance with aspects of the present disclosure.

(117) In related thermal diodes, an extremely high magnetic field is required to achieve, a 6-fold difference in thermal transfer between two different WS particles. Further, the related thermal diodes are required to suspend the WS particles above a substrate in order to fulfill the difference in thermal transfer.

(118) A benefit of a thermal diode system in accordance with aspects of the present example is that a large magnetic field is not required to achieve a large difference in thermal transfer. On the contrary, in accordance with aspects of the present disclosure, because the WS particles are suspended in a dielectric enclosure, there is no need for suspension of the WS particles. Further, a graphene nanograting in conjunction with a rotation of one of the WS particles increases the difference in thermal transfer between two WS particles.

(119) The foregoing description of various preferred embodiments have been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teaching. The example embodiments, as described above, were chosen and described in order to enable others skilled in the art to best utilize the invention in various embodiments and with various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the claims appended hereto.

Claims

1. A thermal diode comprising: a dielectric enclosure having a top surface, a bottom surface, a first side, and a second side; a first Weyl semimetal nanoparticle disposed on the first side; a second Weyl semimetal nanoparticle disposed on the second side; a nanograting disposed on the bottom surface; and a voltage source configured to provide a voltage bias to the nanograting to suppress surface waves from said first Weyl semimetal nanoparticle to said second Weyl semimetal nanoparticle and to modify surface waves from said second Weyl semimetal nanoparticle to said first Weyl semimetal nanoparticle.
2. The thermal diode of claim 1, wherein said first Weyl semimetal nanoparticle and said second Weyl semimetal nanoparticle comprise a material selected from the group of materials consisting of Co.sub.3Sn.sub.2S.sub.2, Mn.sub.3Sn, EuCd.sub.2As.sub.2, TaAs, MoTe.sub.2, NbAs, YbMnBi.sub.2, and WTe.sub.2.
3. The thermal diode of claim 1, wherein said dielectric enclosure comprises SiO.sub.2.
4. The thermal diode of claim 1, wherein the distance between said nanograting and said first Weyl semimetal nanoparticle is 80 nm.
5. The thermal diode of claim 1, wherein said nanograting comprises graphene.
6. A method of making a thermal diode, said method comprising: providing a nanograting; forming a first portion of a dielectric on top of the provided nanograting; depositing a first Weyl semimetal nanoparticle at a first location on top of the first portion of the dielectric; depositing a second Weyl semimetal nanoparticle at a second location on top of the first portion of the dielectric; forming a second portion of the dielectric on top of the first portion of the dielectric, the first Weyl semimetal nanoparticle, and the second Weyl semimetal nanoparticle; and connecting a voltage source to the nanograting, wherein the voltage source is configured to provide a voltage bias to the nanograting to suppress surface waves from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and to modify surface waves from the second Weyl semimetal nanoparticle

to the first Weyl semimetal nanoparticle.

7. The method of claim 6, wherein said depositing the first Weyl semimetal nanoparticle at the first location on top of the first portion of the dielectric first comprises depositing the first Weyl semimetal nanoparticle comprising a material selected from the group of materials consisting of Co.sub.3Sn.sub.2S.sub.2, Mn.sub.3Sn, EuCd.sub.2As.sub.2, TaAs, MoTe.sub.2, NbAs, YbMnBi.sub.2, and WTe.sub.2.

8. The method of claim 6, wherein said forming the first portion of the dielectric on top of the provided nanograting comprises forming the first portion of the dielectric comprising SiO.sub.2.

9. The method of claim 6, wherein said depositing the second Weyl semimetal nanoparticle at the second location on top of the first portion of the dielectric comprises depositing the second Weyl semimetal nanoparticle at from the first Weyl semimetal nanoparticle of 80 nm.

10. The method of claim 6, wherein said providing the nanograting comprises providing a graphene nanograting.

11. A method comprising: providing a thermal diode comprising: a dielectric enclosure having a top surface, a bottom surface, a first side, and a second side; a first Weyl semimetal nanoparticle disposed on the first side; a second Weyl semimetal nanoparticle disposed on the second side; a nanograting disposed on the bottom surface; and a voltage source configured to provide a voltage bias to the nanograting to suppress surface waves from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and to modify surface waves from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle; applying the voltage bias to the nanograting; and rotating the first Weyl semimetal nanoparticle and the second Weyl semimetal nanoparticle about an axis by an angle Θ , wherein $0^\circ \leq \Theta \leq 360^\circ$.

12. The method of claim 11, wherein said providing the thermal diode comprises providing the thermal diode such that the first Weyl semimetal nanoparticle and the second Weyl semimetal nanoparticle comprise a material selected from the group of materials consisting of Co.sub.3Sn.sub.2S.sub.2, Mn.sub.3Sn, EuCd.sub.2As.sub.2, TaAs, MoTe.sub.2, NbAs, YbMnBi.sub.2, and WTe.sub.2.

13. The method of claim 11, wherein said providing the thermal diode comprises providing the thermal diode such that the dielectric enclosure comprises SiO.sub.2.

14. The method of claim 11, wherein said providing the thermal diode comprises providing the thermal diode such that the distance between the nanograting and the first Weyl semimetal nanoparticle is 80 nm.

15. The method of claim 11, wherein said providing the thermal diode comprises providing the thermal diode such that the nanograting comprises graphene.

16. The method of claim 11, further comprising: applying, via the voltage source, a second voltage bias to the nanograting; and rotating the second Weyl semimetal nanoparticle about the axis by a second angle Θ .sub.2, wherein the voltage source is additionally configured to provide the second voltage bias to the nanograting to suppress surface waves from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle and to modify surface waves from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle, wherein a first net heat flux is different from a second net heat flux, wherein the first net heat flux corresponds to a difference between a first heat flux from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and a second heat flux from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle when both Weyl semimetal nanoparticles are rotated about the axis by the angle Θ , wherein the second net heat flux corresponds to a difference between a third heat flux from the first Weyl semimetal nanoparticle to the second Weyl semimetal nanoparticle and a fourth heat flux from the second Weyl semimetal nanoparticle to the first Weyl semimetal nanoparticle when both Weyl semimetal nanoparticles are rotated about the axis by the angle Θ .sub.2, wherein $0^\circ \leq \Theta$.sub.2 $\leq 360^\circ$, and wherein $\Theta \neq \Theta$.sub.2.

17. The method of claim 11, wherein said rotating the first Weyl semimetal nanoparticle and the

second Weyl semimetal nanoparticle about an axis by an angle Θ , comprises applying a magnetic field to both Weyl semimetal nanoparticles.

18. The method of claim 17, wherein said applying the magnetic field to both Weyl semimetal nanoparticles comprises applying the magnetic field to both Weyl semimetal nanoparticles for a predetermined time period.

19. The method of claim 18, wherein said applying the magnetic field to both Weyl semimetal nanoparticles comprises applying the magnetic field at a predetermined field strength to both Weyl semimetal nanoparticles.

20. The method of claim 17, wherein said applying the magnetic field to both Weyl semimetal nanoparticles comprises applying the magnetic field at a predetermined field strength to both Weyl semimetal nanoparticles.
